

Sample plot (Mussoorie ridge)

30.45980°N, 78.06440°E · Dehradun, Dehradun

Report built 2026-10-08 · data build checked 2026-10-08 23:32 · patch 883d102997ffff

The answer

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RELATIVE LANDSLIDE RANK Highest fifth elevated relative risk · percentile 98	LEGACY HAZARD BAND Severe rule-based, context only	TERRAIN RULE: GROUND ≤25° 27% of the patch; not shown to be safer	LIVABILITY 72 /100 access, services, land, community, sun
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How to read the rank. It comes from an independent slope-failure model tested on 67 recorded landslides found after the model was frozen. None fell in the lowest-ranked fifth of the block's hill patches and about 78% fell in the highest fifth (20% each by chance). Within a single district it separates event places from similar ground only modestly (AUC 0.66, 95% interval 0.61–0.71). It is a *relative* screen, not a guarantee.

Hazard stack: what applies at your point

LAYER	APPLIES TO	RESULT
Flood basin (river and torrent floors)	your point, 30 m	above the flood basin
Glacial-lake or glacier outburst corridor	your point, 30 m	not on a traced corridor
Observed monsoon inundation (satellite record)	patch	upland patch, not scored
Relative landslide rank	patch	Highest fifth
Seismic zone	region	Zone VI, the highest category: follow the updated earthquake design code

The 30 m flood basin is the base layer for every plot, upland or plain; the other layers sit on top of it. Point rows describe your exact pixel, patch rows the whole ~0.8 km² hexagon.

Among similar ground in this district (same terrain class and elevation band): **Highest fifth** (percentile 98). Experimental: the district-level rank is still awaiting its own held-out confirmation.

Rule-based flags (context). Flags driven by the national landslide-susceptibility map did not separate event places from similar ground in our held-out tests, so they are not used for the rank.

SEVERE	100% of this patch sits in ILSM's high-or-worse landslide-susceptibility class (100% in the very-high class).
INFO	BIS's 2025 seismic map places the entire Himalayan arc in Zone VI (the highest category) -- construction here should follow the updated earthquake design code.

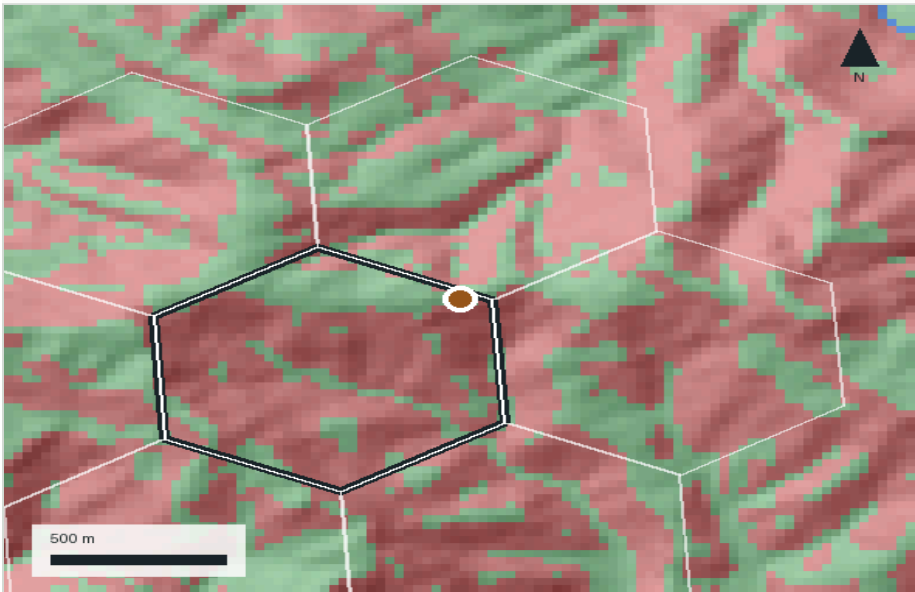
Surrounding patches: 6 Severe — worst nearby: Severe.

Ground at 30 m

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At your point (30 m, not averaged over the patch)

Elevation	2008 m	30 m flood basin at the point	above the flood basin
Position on the slope	ridge or spur	Height above nearest torrent channel	504.7 m · no channel within 1 km
Slope at the point / within 150 m	20.5° / 25.8°	Height above nearest river	1429.4 m · no river within 1 km
Slope faces	south-east (141°) · towards the monsoon flow	Upslope drainage area	0.001 km²
Relief within 300 m	234 m		



- Safe buildable ground
- Too steep (over 25°)
- Torrent or river floor
- Glacial-lake or glacier outburst corridor
- This patch (white inner line); surrounding patches outlined in white
- The point entered

2.6 km square window, 30.9 m pixels, over terrain shading. Classes use the same rules as the safe-buildable share above; the per-patch valley-confinement adjustment is not applied to individual pixels.

Site and confidence

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Coordinates entered	30.45980, 78.06440	Tehsil	Dehradun · 855 patches
Patch	883d102997ffff · 0.785 km²	Safe ground, tehsil average	69%
Point to patch centre	487 m	Legacy hazard bands in the tehsil	Severe 40% · High 17% · Moderate 13% · Low 31%
Terrain	hill · 1,960 m · 28.8°		
Measurements present	14 of 14		

A patch is an H3 hexagon of about 0.74 km². Everything below describes the whole patch containing this point; the map shows what varies inside it.

Mountain hazards in depth

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MEASURED ON THIS PATCH	VALUE	FLAG	RULE APPLIED
Safe buildable ground — 30 m pixels ≤25° and clear of water hazards	27%		Slope ≤25°, ≥5 m above torrents, ≥10 m above rivers, outside corridors
Ground steeper than 35°	22%	no flag	Moderate when 50% of the patch

MEASURED ON THIS PATCH	VALUE	FLAG	RULE APPLIED
Within 30 m height of a river (100 km² catchment)	0%	no flag	High at 10% within 30 m; Severe at 10% within 10 m
Within 5 m height of a torrent (2 km² catchment)	0%	no flag	Moderate at 15%, High at 30%
Valley confinement (a gorge concentrates floods)	1.01×		0.7× wide valley to 1.3× gorge; applied to river and torrent checks
On a traced glacial-lake outburst corridor (Hi-MAG lakes)	0%	no flag	Flagged at 5%; Severe in the near half of the lake's reach
On a traced glacier debris / outburst corridor	0%	no flag	High at 5%
Landslide-dam breach discharge	0.00 m³/s	no flag	High from 50 m³/s, Severe from 150 m³/s
Historical open water (JRC satellite record since 1984)	0%		Corroborates river flood when a share of the patch has been water
100-year river flood depth, global model (WRI Aqueduct)	0.00 m		Context only — about 1 km resolution
Documented disaster incidents on this patch (severity score)	0.0	no flag	Moderate above 0.1, High from 2.0, Severe from 3.5
Cloudburst hotspot rate (20 weather stations, Uttarakhand only)	0.00		Context only — too few stations to describe an area
Legacy, context only: Landslide susceptibility — share of the patch high or very high (ILSM, IIT Delhi)	100%	Severe	Moderate at 20%, High at 50%; Severe when 50% is very high
Legacy, context only: ...of which very high	100%		

Observed monsoon inundation (satellite record)

Upland patch (above about 900 m, not plains or foothill): not scored. River-valley flooding in the hills is covered by the torrent and river rows above.

Access and services

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MEASURE	VALUE
Nearest motorable road	0.0 km
Nearest highway	2.4 km
District capital	26.2 km
Major commercial centre	26.2 km
Nearest hospital	4.2 km
Nearest school	3.6 km
Shops, banks, restaurants within 1 km	9 places
...within 5 km	51 places
Population density	2,159.4 people/km²
Building growth, 2016–2023	5.2 %/yr
Night-light growth, 2014–2025	1.33 ×

Distances in the table above follow the road network. Building growth is left blank where the 2023 building record has a gap.

Road class, electricity and banks (straight-line, OpenStreetMap)

MEASURE	VALUE
Nearest major road (national / state highway class)	0.4 km
Nearest secondary road	6.1 km
Nearest tertiary road	0.3 km
Nearest local road or track	0.0 km
Road length inside the patch: major / secondary / tertiary / local	1.9 / 0.0 / 3.8 / 29.8 km
Nearest mapped power line (OSM, transmission and distribution)	13.7 km
Nearest mapped substation (OSM)	11.5 km
Nearest bank or ATM	1.2 km
Banks and ATMs within 5 km	10

Straight-line distances from the patch centre, not travel time. OpenStreetMap is patchy for minor roads and power lines in remote hills, so a missing feature may be unmapped, not absent. Whether a village is electrified, and the quality of supply, are not in open data and should be checked locally.

Climate, weather and water6

30 measured rows for this patch, in the grammar global asset-risk reports use so the two can be read side by side. Each row shows the value in its own unit and a fixed 1–5 score (1 = low, 5 = very high); thresholds are fixed, so scores are comparable between sites. These are context for the site and for planning, not predictions of events.

HAZARD	TYPE	VALUE	UNIT	SCORE 1–5
CONTEXT				
Worldcover 2020	context	tree	class	–
World Settlement Footprint Evolution	context	0.01	share	–
SEISMIC				
Earthquake	zone	VI	BIS zone	5
Soil & Shaking	zone	624.3	m/s	3
Distance to Active Faults	zone	21.86	km	3
SOLID MASS				
Avalanche	acute	0.0	share	1
Subsidence	acute	26.7	%	–
Soil Erosion	chronic	36.72	t/ha/yr	4
Soil Degradation	chronic	2	processes (0-5)	3
Solifluction	chronic	66.3	days/yr	3
TEMPERATURE				
Heat Wave	acute	1.67	days/yr	2
Cold-Frost	acute	66.3	days/yr	3

HAZARD	TYPE	VALUE	UNIT	SCORE 1-5
Wildfire	acute	0.0	% of patch/yr	1
Fire Weather Stress Index	acute	22.08	FWI	3
Changing Temperature	chronic	0.07	C	—
Heat Stress	chronic	29.46	C	1
Heat-Humidity Stress Index	chronic	18.09	C	1
Temperature Variability	chronic	10.9	C	2
Permafrost Thawing	chronic	0.0	% of area	1
WATER				
Hail	acute	0.8	days/yr	3
Drought	acute	10.28	% of months	4
Heavy Precipitation	acute	116.58	mm	3
Flood	acute	1	1-5	1
River Flood (Defended)	acute	—	years	1
Changing Precipitation Patterns	chronic	0.46	%	—
Precipitation Variability	chronic	0.82	index	4
Water Stress	chronic	4	Aqueduct category	5
Water Scarcity	chronic	3	Aqueduct category	4
WIND				
Storm	acute	84.44	km/h	2
Changing Wind Patterns	chronic	-2.65	%	—

Long-run trends

Warming trend (ERA5, full record)0.0167 °C/yr

Monsoon rainfall variability (CHIRPS, coefficient of variation)0.378

Vegetation health (MODIS NDVI, annual mean)0.544

A 10-year outlook (warming-trend based) is in development and is not part of this report.

Can I buy, can I build

CHECK	STATUS	DETAIL
Seismic design	disclosed	BIS's 2025 seismic map places the whole Himalayan arc in Zone VI, the highest category — build to the updated earthquake code.
Who may buy this land	not assessed yet	Notified rules to be checked against the gazette; not part of this report.
River setbacks	not assessed yet	Distance rules from rivers; not part of this report.
Eco-sensitive zones	not assessed yet	Notified zone boundaries; not part of this report.

How good is this

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The data build behind this report passed its automated checks on 2026-10-08 23:32 (PASS, build score 99, 0 failed checks).

What was tested. The landslide rank was trained on the Geological Survey of India landslide inventory (Himachal) and tested on 67 recorded landslides from news and other reports found after the model was frozen. Across the block, none fell in the lowest-ranked fifth of hill patches and about 78% in the highest fifth. Against similar ground in the same district the AUC is 0.66 (95% interval 0.61–0.71; 0.5 is chance), and one batch of 18 events scored lower (0.56). So the rank is strong at the regional level and modest for telling one plot from its neighbours. Samples are small and news reports favour roads and towns, so read these as direction, not exact rates.

What did not hold up. The national landslide-susceptibility map and the rule-based “Severe” band showed almost no skill on independent events; “safe buildable ground” (slope $\leq 25^\circ$) was not shown to be safer; river proximity is a *risk marker* (events cluster beside streams), not a safety signal.

Not in this report yet

- Climate projections by scenario and year
- Ground subsidence from radar
- Land purchase, zoning and building rules (the diligence note covers what to check)
- Structural and earthquake-engineering assessment
- Electrification status and quality of supply
- Floods after 2021 in the observed-inundation record

Known limits of the method

- The landslide rank is relative: it orders patches against similar ground, it does not give a probability or a guarantee, and within one district it separates plots only modestly.
- Flood-basin classes use fixed height-above-channel rules at 30 m; they are not hydraulic flood models and give no depth or timing.
- Observed monsoon inundation covers low-lying ground for 1984 to 2021; narrow gorges and floods that left no cloud-free satellite scene are missed, and floods after 2021 are not in it.
- Short, intense cloudburst rain is not a modelled input: satellite rainfall grids of 5 to 11 km cannot resolve a cloudburst cell.
- Glacial-lake and glacier outburst corridors are traced from mapped lakes and glaciers and cut off at each data block's edge.
- Roads, power lines and banks come from OpenStreetMap, which is patchy for minor roads and power lines in remote hills; distances are straight-line.
- Landslide-dam breach discharge is modelled for the West block (Uttarakhand west of the Alaknanda) only.

Method and sources

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Each patch is a hexagon of about 0.8 km². Terrain, slope and drainage come from the Copernicus 30 m elevation model; flood basins from height above the nearest river or torrent channel; outburst corridors are traced downstream from mapped glacial lakes and glaciers. The landslide rank is a model trained on the Geological Survey of India landslide inventory (Himachal) and tested on recorded landslides found afterwards. Observed monsoon inundation is the share of July–August Landsat observations (1984–2021) showing water, graded against independent flood maps. Every threshold lives in one rules file with its reason, and every figure in this report is stored in `figures.json` beside it.

SOURCE	USED FOR	TERMS
Copernicus DEM GLO-30	Elevation, slope, drainage, ground map	Copernicus DEM licence
ILSM landslide susceptibility (IIT Delhi)	Landslide share	Research dataset — see source
Hi-MAG glacial lakes (Chen et al. 2021)	Outburst corridors	CC BY 4.0
Randolph Glacier Inventory	Glacier debris corridors	See source
JRC Global Surface Water (monthly history)	Historical open water; observed monsoon inundation	Copernicus / JRC terms
Global Flood Database (Cloud to Street)	Calibration of inundation grades	CC BY 4.0
Geological Survey of India landslide inventory (Himachal subset)	Training of the landslide rank	CC BY 4.0 (Mendeley Data redistribution)
News and agency reports of landslides, floods and cloudbursts	Independent testing of the landslide rank	Per-article sources kept on file
WRI Aqueduct Floods	100-year flood depth (context)	See source
OpenStreetMap	Roads, services, shops	ODbL
Google Open Buildings temporal	Building growth	CC BY 4.0
WorldPop · VIIRS night lights · ERA5 · CHIRPS · MODIS	Population, activity, climate context	See each source
Documented incidents compilation	Incident flags	GSI, NIDM, NDMA, USDMA records

A screening report, not a site survey, engineering assessment or legal opinion. Visit the site and consult a qualified engineer before buying or building. Figures: figures.json · Patch 883d102997ffff · Schema hill-site-report/v1